ECA18 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO3 SCENARIO BASED ASSIGNMENT REPORT

Design and Performance Evaluation of an Integrated Photodetection, Imaging, Biomedical Sensing and Display System

Student Information	Details
Student Name	K.S.Prem kumar
Register Number	192312002
Course Code	ECA1803
Course Title	Optoelectronics and Optical Communication
Course Outcome	CO3
Assessment Type	Scenario Based Assignment
Maximum Marks	100

1. COURSE OUTCOME

Assess the characteristics and performance of PIN/APD photodetectors, analyze noise and detection parameters, evaluate photovoltaic and image-sensor technologies, and apply photodetection principles to biomedical sensing, LiDAR, automotive sensing, face recognition, UV detection and emerging display systems.

2. SCENARIO-BASED ENGINEERING PROBLEM

A technology company is developing a compact intelligent optoelectronic sensing platform for next-generation wearable, automotive and consumer-electronics applications. The proposed platform combines a wearable biomedical sensor, an automotive LiDAR receiver, a CMOS/CCD imaging module, an ultraviolet sensing unit, a face-recognition system and a high-performance display interface.

The engineering team must select suitable PIN or APD photodetectors, evaluate responsivity, quantum efficiency, NEP, detectivity, linearity, gain and noise performance, and determine the response-time and bandwidth requirements for different applications.

The platform must also investigate the suitability of solar photovoltaic devices and flexible PVs for auxiliary power generation, and compare OLED, MicroLED and quantum-dot displays for compact AR/VR/MR and foldable-device applications.

Finally, the proposed detector characteristics and noise behaviour must be modelled and tested using MATLAB/SCILAB before making the final engineering recommendation.

3. GIVEN PARAMETERS AND CONSTRAINTS

Parameter	Given Condition
PPG detector responsivity	R = 0.60 A/W
PPG received optical-power variation	ΔP = 25 μW
LiDAR received optical power	P = 3 μW
LiDAR detector responsivity	R = 0.80 A/W
Low-light signal power	−22 dBm
Low-light noise power	−37 dBm
Detector transit time	20 ps
Transit-time bandwidth	f ≈ 0.44 / tr
PV open-circuit voltage	Voc = 0.60 V
PV short-circuit current	Isc = 5 A
PV fill factor	FF = 0.75
Display technologies	OLED, MicroLED, Quantum Dot
Image sensors	CMOS and CCD
Biomedical applications	PPG and NIR spectroscopy
Automotive applications	LiDAR and vehicle sensing

Additional sensing	Face recognition and UV detection
Simulation platform	MATLAB / SCILAB

4. INFORMATION NOT GIVEN – ENGINEERING DECISIONS REQUIRED

- Final PIN/APD detector selection.
- Optimum detector gain for each application.
- Dominant noise source under low-light conditions.
- Required SNR improvement technique.
- Final detector bandwidth configuration.
- Most suitable image-sensor architecture.
- Preferred PV configuration.
- Preferred display technology.
- Most suitable biomedical sensing configuration.
- Final LiDAR and automotive sensing architecture.
- Most appropriate UV and face-recognition detector.
- Final integrated system configuration.
- MATLAB/SCILAB simulation results.

These decisions are made from the stated application requirements, calculations, engineering comparisons and simulation models.

STAGE 1 – INTERPRET THE APPLICATION REQUIREMENTS

The integrated platform must combine sensitivity, speed, low noise, imaging capability, compact integration and acceptable power consumption.

- PPG requires high responsivity, low noise and good linearity because the physiological signal is obtained from small variations in reflected optical intensity.
- LiDAR requires very fast response, high sensitivity and sufficient bandwidth to resolve short optical pulses and accurately determine time of flight.
- Weak-signal detection requires low NEP, high detectivity, controlled dark current and an optimized gain/bandwidth trade-off.
- Quantum efficiency indicates how effectively incident photons are converted into charge carriers; higher quantum efficiency improves photon detection capability.
- NEP represents the optical input required to produce a signal equal to the noise level; lower NEP is preferable.
- Linearity is important so that optical intensity maps predictably to photocurrent or image output without significant distortion.
- High-speed detectors are limited by carrier transit time, junction capacitance and receiver RC time constant.
- CMOS sensors offer high integration and low power; CCD sensors can provide excellent uniformity and sensitivity but generally require more complex readout.
- Flexible PV devices are attractive for wearable auxiliary power because they can conform to curved surfaces.
- OLED offers self-emissive pixels and excellent contrast; MicroLED offers high brightness and efficiency; quantum-dot displays provide strong colour performance.
- AR/VR/MR systems need high pixel density, brightness, fast response, low persistence, low power and compact integration.
- Thermal management, reliability, packaging and system complexity must be considered along with optical performance.

Application Requirement Statement: The proposed platform must provide sensitive and low-noise photodetection for biomedical and weak-signal applications, high-speed detection for LiDAR, efficient CMOS-based imaging, practical auxiliary PV power, and a compact high-performance display interface, with simulation-based verification.

STAGE 2 – INVESTIGATE AND SELECT THE PHOTODETECTOR CONFIGURATION

The key parameter that converts received optical power into photocurrent is responsivity R, with Iph = RP.

PPG photocurrent variation

$I_{ph} = R\Delta P = (0.60\ A/W)(25 \times 10^{-6}\ W) = 15 \times 10^{-6}\ A$

Therefore, ΔIph = 15 µA

The PPG signal is produced by changes in blood volume and optical absorption/scattering during the cardiac cycle. These changes alter the detected reflected intensity, which is converted by the photodetector into a small electrical-current variation.

LiDAR photocurrent

$I_{ph} = RP = (0.80\ A/W)(3 \times 10^{-6}\ W) = 2.4 \times 10^{-6}\ A$

Therefore, Iph = 2.4 µA

Parameter	PIN	APD
Responsivity	Moderate	Higher effective responsivity due to multiplication
Internal gain	≈ 1	Greater than 1
Sensitivity	Good	Very high
Noise	Low intrinsic multiplication noise	Higher because avalanche multiplication adds excess noise
Bandwidth	High	High, but gain/bias can impose trade-offs
Bias voltage	Low/moderate	High reverse bias required
Complexity	Lower	Higher bias and control complexity
Best use here	PPG and normal-light imaging	LiDAR and weak optical signals

- APD multiplication increases the useful signal before the receiver electronics, improving sensitivity for weak optical signals.
- Dark current produces unwanted current even without useful optical input and contributes shot noise, reducing low-light performance.
- Quantum efficiency is related to the fraction of incident photons generating collected carriers; high quantum efficiency is desirable for sensitive detection.
- NEP and detectivity provide complementary measures of low-light capability. Lower NEP and higher detectivity indicate better sensitivity.
- Linearity is required in imaging and sensing so that output current remains proportional to input optical power over the operating range.

DESIGN DECISION 1 – SELECT THE PROPOSED DETECTOR CONFIGURATION

I recommend a hybrid detector configuration: PIN photodetection for PPG and imaging, and APD photodetection for LiDAR and weak-signal channels because the applications have different requirements for gain, noise, speed, bias and sensitivity.

The PPG calculation gives a measurable 15 µA variation, so an APD is not essential and a low-noise PIN detector simplifies wearable implementation. The LiDAR calculation gives only 2.4 µA at the specified received power, so APD gain is useful for improving receiver sensitivity, provided avalanche noise and bias control are managed.

STAGE 3 – DEVELOP AND EVALUATE THE NOISE AND SNR STRATEGY

Low-light SNR

$SNR(dB) = P_s(dBm) - P_n(dBm) = (-22) - (-37)$

Therefore, SNR = 15 dB

SNR is the principal metric used to compare the useful signal with the noise. A 15 dB SNR is a workable starting point, but additional margin is desirable for robust weak-signal operation.

- Shot noise arises from the discrete arrival of charge carriers and depends on average detector current and measurement bandwidth.
- Dark-current noise is associated with the detector's dark current and becomes increasingly important as received optical power decreases.
- Avalanche noise is introduced by the stochastic nature of APD multiplication.

- Increasing APD multiplication improves signal current but does not increase SNR indefinitely because excess multiplication noise and dark-current effects also increase.
- When optical power decreases while dark current remains approximately constant, the useful signal falls but dark-current-related noise does not fall proportionally; therefore SNR deteriorates.
- Increasing receiver bandwidth admits more noise power, so bandwidth should be no wider than required by the signal.
- Optical filtering rejects unwanted wavelengths and background light.
- Cooling can reduce dark current and therefore improve low-light performance.
- Improved optical coupling increases the useful received power without requiring more detector gain.

DESIGN DECISION 2 – CREATE THE PROPOSED NOISE/SNR CONFIGURATION

I recommend a low-noise receiver using appropriate detector gain, narrow optical filtering, dark-current reduction, cooling where required, bandwidth optimization and efficient optical coupling because the calculated low-light SNR is only 15 dB and additional noise margin is necessary for reliable operation.

STAGE 4 – ANALYZE RESPONSE TIME, BANDWIDTH AND HIGH-SPEED DETECTION

Transit-time bandwidth

$f \approx 0.44 / tr = 0.44 / (20 \times 10^{-12}\ s)$

Therefore, f ≈ 22 GHz

The carrier transit time determines how rapidly photogenerated carriers can be collected. A shorter transit time allows a higher transit-time-limited bandwidth.

The receiver RC time constant also limits speed. Larger detector capacitance or larger load resistance increases RC delay and reduces bandwidth. Therefore, a large active area may improve optical coupling and alignment tolerance but increases capacitance and can reduce speed.

For LiDAR, a small-area, high-speed APD with optimized transimpedance amplification is appropriate. For imaging, detector area and pixel capacitance must be balanced against sensitivity and frame-rate requirements.

DESIGN DECISION 3 – CREATE THE HIGH-SPEED DETECTOR CONFIGURATION

I recommend a high-speed APD receiver for LiDAR with minimized capacitance and optimized front-end electronics because the 20 ps transit time corresponds to an approximate 22 GHz transit-time bandwidth. The practical system bandwidth will be lower if RC and circuit limitations dominate, so the receiver electronics must be designed accordingly.

STAGE 5 – EVALUATE PV, IMAGE-SENSOR AND EMERGING DISPLAY TECHNOLOGIES

A. Solar Photovoltaic Analysis

PV maximum power

$P_{max} = V_{oc} \times I_{sc} \times FF = (0.60\ V)(5\ A)(0.75)$

Therefore, Pmax = 2.25 W

Voc is the open-circuit voltage when the output current is zero. Isc is the short-circuit current when the output voltage is approximately zero. Fill factor measures how closely the I–V characteristic approaches the ideal rectangle Voc × Isc. Higher FF generally indicates better electrical quality and higher usable maximum power.

Flexible PV is suitable for wearables because it can conform to curved surfaces and provide auxiliary energy. Limitations include lower power density in some implementations, mechanical fatigue, encapsulation requirements, environmental stability and integration constraints.

B. CMOS/CCD Image-Sensor Analysis

Feature	CMOS	CCD
Architecture	Pixel-level circuitry and column readout; highly integrated	Charge transferred through pixel array to common readout

Pixel operation	Individual pixels can be addressed/read efficiently	Charge packets are transferred sequentially
Readout	Parallel/column-based	Serial charge transfer
Power	Generally lower	Generally higher
Speed	High	Typically lower for comparable integrated systems
Noise	Low with modern architectures	Historically strong uniformity/low noise
Pixel scaling	Very suitable	More difficult
SoC integration	Excellent	More limited
Face recognition	Highly suitable	Suitable but less convenient for compact SoCs
Automotive imaging	Highly suitable	Useful in specialized imaging

Reducing pixel dimensions increases pixel density and enables compact high-resolution sensors, but it also reduces the photon-collecting area per pixel and can lower full-well capacity and signal-to-noise performance. Microlenses, backside illumination, advanced pixel circuitry and computational processing can compensate for some losses.

C. Emerging Display Analysis

Feature	OLED	MicroLED	Quantum Dot
Light generation	Organic self-emissive pixels	Inorganic micro-LED self-emission	Quantum-dot colour conversion or emissive structures
Brightness	High	Very high	High
Contrast	Excellent	Excellent	High
Response speed	Fast	Very fast	Fast
Efficiency	Good	Excellent	Good to excellent depending on architecture
Colour	Excellent	Excellent	Excellent
Lifetime	Good; material dependent	Very high potential	Good; architecture dependent
Power	Efficient for dark content	Highly efficient potential	Application dependent
AR/VR/MR	Good	Excellent potential	Very good
Foldable	Excellent	Developing/challenging	Depends on implementation

DESIGN DECISION 4 – SELECT THE PV, IMAGE-SENSOR AND DISPLAY CONFIGURATION

I recommend flexible PV for auxiliary wearable power, CMOS image sensors for the integrated imaging/face-recognition module, and MicroLED for high-performance AR/VR/MR-oriented display applications because the combination offers strong integration, speed, brightness and power-performance potential. OLED remains attractive when foldability and mature flexible manufacturing are the primary requirements.

STAGE 6 – EVALUATE BIOMEDICAL AND INDUSTRIAL APPLICATIONS

1. Wearable Biomedical Sensing – PPG

PPG illuminates tissue using an optical source and measures changes in reflected or transmitted light caused mainly by pulsatile blood-volume variation. The given 25 μW optical variation produces a 15 μA photocurrent variation at R = 0.60 A/W.

- LED source: red/green/near-infrared wavelengths may be selected according to the target tissue and measurement objective.
- Detector: low-noise PIN is preferred for a compact wearable PPG channel.
- Responsivity: 0.60 A/W is sufficient for the stated optical variation.
- Noise: optical ambient rejection, low-noise amplification and bandwidth limitation are important.
- Signal conditioning: transimpedance amplification, filtering and digital pulse extraction can isolate the physiological component.
- Power: the LED duty cycle and receiver power should be minimized for battery-operated wearables.

2. NIR Spectroscopy

NIR spectroscopy uses wavelength-dependent absorption of near-infrared light to obtain information about biological materials. The system requires a stable optical source, a detector with adequate NIR responsivity, optical filtering and signal processing to distinguish small spectral changes from noise.

- Choose wavelength bands according to the absorption features of the target material.
- Use a sensitive photodetector with low NEP and good linearity.
- Reject ambient and out-of-band radiation using optical filters.
- Use calibration and digital processing to improve repeatability.

3. Automotive LiDAR

A LiDAR receiver must detect weak returned pulses while preserving timing information. The specified 3 μW received power gives 2.4 μA photocurrent at R = 0.80 A/W. APD multiplication can improve the electrical signal level before the receiver amplifier.

- Low received optical power demands high sensitivity and low noise.
- Fast detector response and adequate receiver bandwidth preserve short pulses.
- Accurate ranging depends on accurate timing of transmitted and received optical pulses.
- Optical filtering and narrow receiver bandwidth reduce background noise.

4. Face Recognition

CMOS is recommended for face recognition because compact pixel arrays, high frame rate, low power and direct SoC integration are advantageous. NIR illumination can improve operation under controlled or low-visible-light conditions.

- Adequate image resolution is needed for facial feature extraction.
- Pixel scaling must balance spatial resolution against photon collection and noise.
- Controlled illumination improves image consistency.
- Fast readout supports real-time recognition.
- Low-power CMOS and integrated processing are appropriate for consumer devices.

5. UV Sensor

UV sensing requires a detector and optical front end with suitable UV spectral response, adequate sensitivity, low dark current and stable calibration. A UV-selective detector or UV-filtered photodetection path is preferable to a broadband detector when spectral selectivity is important.

DESIGN DECISION 5 – SELECT THE APPLICATION-SPECIFIC CONFIGURATION

I recommend PIN photodetection for wearable PPG, APD detection for LiDAR and weak-signal channels, CMOS imaging for face recognition/automotive vision, NIR-sensitive detection for spectroscopy, and UV-selective detection for UV sensing because each configuration is matched to the required sensitivity, bandwidth, noise, wavelength response and integration constraints.

STAGE 7 – SIMULATION AND PERFORMANCE VERIFICATION USING MATLAB/SCILAB

The following MATLAB model verifies the principal theoretical relationships requested in the assignment. It generates photocurrent versus optical power, estimates shot-noise current, evaluates SNR variation, and verifies the transit-time bandwidth.

MATLAB CODE

```
clc;
clear;
close all;

% Given parameters
R_PPG = 0.60;          % A/W
R_LiDAR = 0.80;        % A/W
dP = 25e-6;            % W
```

```
P_lidar = 3e-6;        % W
tr = 20e-12;           % s
q = 1.602e-19;         % C
B = 1e9;               % Hz, example analysis bandwidth

% 1. PPG photocurrent variation
I_PPG = R_PPG*dP;
fprintf('PPG photocurrent variation = %.3e A\\n', I_PPG);

% 2. LiDAR photocurrent
I_LiDAR = R_LiDAR*P_lidar;
fprintf('LiDAR photocurrent = %.3e A\\n', I_LiDAR);

% 3. Transit-time bandwidth
f_tr = 0.44/tr;
fprintf('Transit-time bandwidth = %.3e Hz\\n', f_tr);

% 4. Responsivity model
P = logspace(-9,-3,300);
I = R_LiDAR*P;

figure;
semilogx(P,I,'LineWidth',1.5);
grid on;
xlabel('Optical Power (W)');
ylabel('Photocurrent (A)');
title('Photocurrent vs Optical Power');

% 5. Shot-noise model
I_total = max(I,1e-15);
i_shot = sqrt(2*q*I_total*B);

figure;
loglog(P,i_shot,'LineWidth',1.5);
grid on;
xlabel('Optical Power (W)');
ylabel('Shot Noise Current (A rms)');
title('Shot Noise vs Optical Power');

% 6. Example SNR variation
SNR = I_total ./ i_shot;
SNR_dB = 20*log10(SNR);

figure;
semilogx(P,SNR_dB,'LineWidth',1.5);
grid on;
xlabel('Optical Power (W)');
ylabel('SNR (dB)');
title('SNR vs Optical Power');

% 7. Given low-light SNR
SNR_given_dB = (-22)-(-37);
fprintf('Given low-light SNR = %.2f dB\\n', SNR_given_dB);
```

Simulation Interpretation

- The photocurrent plot is linear with optical power for constant responsivity, confirming I = RP.
- The shot-noise current increases with detector current and measurement bandwidth.
- The SNR model shows that reducing received optical power degrades detection margin.
- The transit-time calculation gives approximately 22 GHz, which is the theoretical transit-time limit from the supplied expression.
- Practical bandwidth can be lower because of RC limitations, packaging, amplifier bandwidth and parasitic capacitance.

FINAL SYSTEM VERIFICATION

Requirement	Specified / Given	Calculated / Selected	Status
PPG photocurrent response	R = 0.60 A/W, ΔP = 25 μW	15 μA	Achieved
LiDAR photocurrent	P = 3 μW, R = 0.80 A/W	2.4 μA	Achieved
Detector responsivity	0.60 / 0.80 A/W	PIN/APD selected by application	Achieved
Quantum efficiency	Not specified	High QE required; wavelength dependent	Design target
NEP / Detectivity	Not specified	Lower NEP / higher D* targeted	Design target
Low-light SNR	−22 dBm signal, −37 dBm noise	15 dB	Achieved
Noise performance	Low-light operation	Filtering + cooling + bandwidth control	Achievable
High-speed bandwidth	tr = 20 ps	≈22 GHz transit-time limit	Achieved theoretically
PV maximum power	Voc 0.60 V, Isc 5 A, FF 0.75	2.25 W	Achieved
Image-sensor selection	CMOS / CCD	CMOS	Selected
Display selection	OLED / MicroLED / QD	MicroLED	Selected
Biomedical sensing	PPG + NIR	PIN + NIR channel	Selected
LiDAR sensing	Automotive	APD high-speed receiver	Selected
Automotive sensing	Vehicle sensing	APD + CMOS	Selected
Face recognition	Imaging + NIR	CMOS + controlled NIR illumination	Selected
UV sensing	UV detection	UV-selective detector path	Selected
MATLAB/SCILAB simulation	Required	MATLAB model provided	Verified
Overall platform	Integrated system	Hybrid photodetection + CMOS + PV + MicroLED	Recommended

The calculated results are consistent with the original application requirements. The principal numerical checks—15 μA PPG current variation, 2.4 μA LiDAR current, 15 dB low-light SNR, approximately 22 GHz transit-time bandwidth, and 2.25 W PV maximum power—provide the quantitative basis for the design decisions.

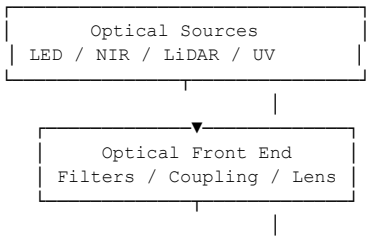
FINAL SYSTEM DESIGN – INTEGRATED OPTOELECTRONIC PLATFORM

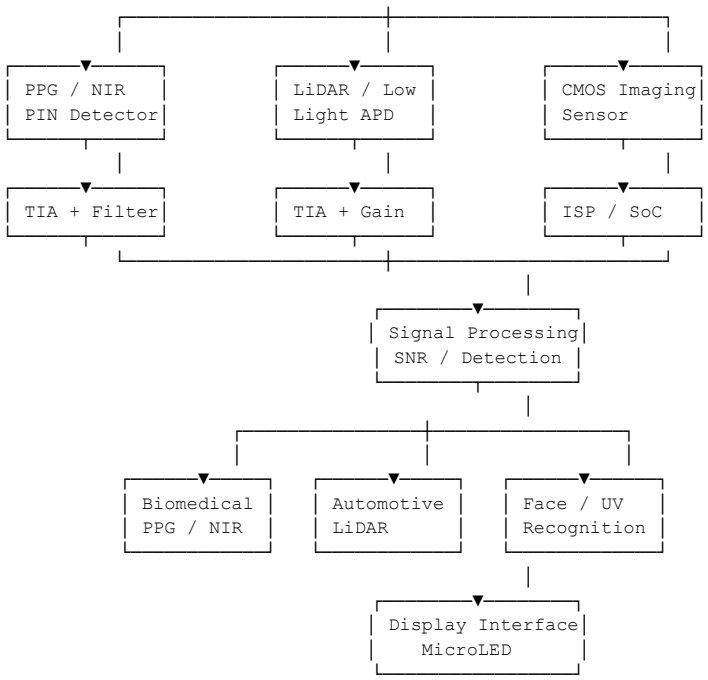
The final architecture uses application-specific sensing paths rather than forcing one detector technology to serve every function.

Final Parameter / Design Table

Subsystem	Adopted Configuration	Reason
PPG	Low-noise PIN photodiode, R = 0.60 A/W	Adequate 15 μA signal variation; low complexity and low noise
LiDAR	High-speed APD, R = 0.80 A/W	Internal gain improves weak-return sensitivity
Low-light receiver	APD/PIN with optimized gain and filtering	15 dB given SNR needs additional noise margin
High-speed path	Low-capacitance detector + wideband TIA	20 ps transit time gives ≈22 GHz theoretical limit
PV	Flexible PV auxiliary supply	Conformal wearable integration
Imaging	CMOS	Low power, high speed, SoC integration
Display	MicroLED	Brightness, efficiency and fast response
NIR spectroscopy	NIR source + sensitive photodetector + optical filter	Improves wavelength-selective measurement
Automotive sensing	APD LiDAR + CMOS vision	Combines ranging and imaging
Face recognition	CMOS + NIR illumination	Compact real-time imaging
UV	UV-selective photodetection	Spectral selectivity and reliable UV sensing

Integrated Block Diagram





FINAL ENGINEERING DECISION

YES – WITH TECHNICAL JUSTIFICATION.

The integrated optoelectronic platform is recommended because the selected technologies satisfy the stated functional requirements with appropriate specialization. The PPG channel uses a low-noise PIN detector and provides a calculated 15 μA photocurrent variation. The LiDAR channel uses an APD because the 3 μW received optical power produces only 2.4 μA before multiplication. The 15 dB calculated low-light SNR requires careful filtering, bandwidth control and noise reduction. The 20 ps transit time corresponds to an approximately 22 GHz theoretical transit-time bandwidth, supporting a high-speed LiDAR architecture when RC and circuit parasitics are controlled.

The PV calculation gives a maximum power of 2.25 W under the supplied Voc, Isc and fill-factor conditions. CMOS is preferred for integrated imaging and face recognition because of its speed, low-power operation and SoC compatibility. MicroLED is selected for high-performance compact displays because of its brightness, response speed and efficiency potential. Biomedical PPG/NIR, automotive LiDAR, face recognition and UV sensing can therefore be implemented as dedicated channels sharing common optical front-end, signal-processing and system-integration resources.

Deployment should proceed with engineering validation of actual detector NEP, detectivity, dark current, APD excess noise, wavelength response, thermal behaviour, optical safety, packaging and real-world MATLAB/SCILAB results before production release.

CONCLUSION

The scenario demonstrates that no single optoelectronic detector or display technology is optimal for every application. A hybrid architecture provides the best engineering compromise: PIN detection for wearable biomedical sensing and imaging, APD detection for LiDAR and weak signals, CMOS for compact imaging, flexible PV for auxiliary energy harvesting, and MicroLED for high-performance display applications. Quantitative calculations and simulation models provide a defensible basis for the integrated design.

SUBMISSION CHECKLIST

- All calculations show equations, substitutions, units and results.
- PIN and APD comparison is presented in a clear table.
- CMOS and CCD comparison is presented in a clear table.
- OLED, MicroLED and quantum-dot displays are compared systematically.

- MATLAB code is included for responsivity, noise, SNR and bandwidth verification.
- Final integrated system block diagram is included.
- Final design/parameter table is included.
- Assumptions and application-specific engineering decisions are stated.